



Sustainable Development Verified Impact Standard

IMPROVED COOKSTOVE GROUPED PROJECT IN PAPUA NEW GUINEA



Document Prepared by TEM PNG Cookstoves Project Pte Ltd

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Project Lifetime	23 September 2024 to 22 September 2034; 10-year lifetime
Monitoring Period of this Report	23 September 2024 – 31 December 2024
History of SD VSta Status	SD VSta Registered on 14/06/2024 ¹
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¹ <https://registry.verra.org/app/projectDetail/SDVISTA/4416>

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1 SUMMARY OF SDG CONTRIBUTIONS

Table 1: Summary of SDG Contributions

Row number	Quantitative Project Contributions during Monitoring Period	Contributions during Project Lifetime	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Section Reference	Claim, Asset or Label
1)	For this current monitoring period, a total of 19,206 traditional cookstoves were replaced with fuel efficient Improved Cookstoves (ICSs) in the project area, with a focus in the Southern Highlands Province. The savings per household per day would be 1.04² Kina for households that purchase their own firewood. By proportioning the number of households that purchase their firewood from the monitoring survey, 57,890 Kina⁷ is saved	The efficiency of changing from a traditional cookstove to a more efficient one will result in the consumption of lesser firewood that is usually purchased at high prices,³ hence reducing the economic burden of families that buy firewood under the project activity. The project activity has replaced 19,206 traditional cookstoves with fuel efficient ICSs, leading to savings of	1.1	Goal 1: Amount of money saved by families that buy firewood	Implement activities to increase	3.2 #1	Claim

² The calculation method is demonstrated in the SD VISTa Monitoring Plan Spreadsheet made available to validators.

³ ACIAR 2013_Promoting diverse fuelwood production systems in Papua New Guinea

	during this period across all households.	1.04 Kina per household per day. By proportioning the number of households that purchase their firewood from the monitoring survey, 57,890 Kina ⁷ is saved during this period.					
2)	Emissions of PM2.5 were estimated to be reduced to below the baseline emission level by 81.24% to 177 mg/MJ⁴ in project households for newly distributed ICSs, based on estimated from literature for the same stove model. For ICSs distributed for more than a year, there would be a decrease in thermal efficiency, however, for this current monitoring period, no ICSs were more than a year old, as all 19,206 ICSs were newly distributed.	Contribute to improved health and well-being brought about by reduced levels of fine particulate matter (PM_{2.5}) emissions within households below the baseline emission level of 943 mg/MJ delivered⁶. Overall, a total decrease of 244 tonnes of PM2.5⁷ has been observed during the project lifetime.	3.9	Goal 3: Reduction of fine particulate matter (PM2.5) emissions within households to reduce air pollution within households in the project area.	Implemented activities to increase	3.2 #2	Claim

⁴ http://catalog.cleancookstoves.org/stove_details.html?stove_id=stove_AW0UJ

⁶ http://catalog.cleancookstoves.org/stove_details.html?stove_id=stove_A8WGMJX

⁷ The calculation method is demonstrated in the SD VISTA Monitoring Plan Spreadsheet made available to validators.

	Overall, this has resulted in a total decrease of 244.79 tonnes⁵ of PM2.5 during this monitoring period based on the theoretical reduction of baseline emissions due to the use of ICSs² as compared to traditional cookstoves³. Additionally, this is further supported by the Ground Monitoring Survey which reported a 100% of households have observed a decrease in intensity of the smoke compared to the baseline scenario.						
3)	Non-formal education in the form of vocational and relevant skills was provided to 24 local individuals with respect to issues related to climate change, with specific skill building in operations and surveying activities related to ICS distribution and its monitoring under VCS have been provided during this monitoring period. All staff met	Contributed to increasing vocational and relevant skills of local individuals by providing non-formal education and training on issues related to climate change, with specific skill building in operations and surveying activities related to ICS distribution and its monitoring under VCS.	4.3	Goal 4.1: Number of youth and adults in formal and non-formal education and training in the previous 12 months, by sex	Increase	3.2 #5	Claim

⁵ The calculation method is demonstrated in the SD VISTA Monitoring Plan Spreadsheet made available to validators.

	the minimum passing rate of the quizzes given to them.	During the project lifetime, a total of 24 local staff were trained in training courses. These training courses were conducted annually for all the staff to make sure that the knowledge transferred is up to date and the delivery of the project remains on a high standard. At the end of each training session, staff sat for an informal quiz to test their understanding of the training material. The measurement of improvement of learning outcomes and skill development will be based on the passing rate of the quiz. The minimum passing rate is at least 70%, and if a staff falls below the mark, they will have to re-sit the quiz within the same day.					
4)	During this monitoring period, a total of 24 local staff were trained to conduct regular monitoring of the project sites for additional monitoring and maintenance of the ICSs and	Contributed to increasing the number of youth and adults who have relevant skills, including technical and vocational skills, for	4.4	Goal 4.2: Number of youth and adults with technical and vocational skills, by type	Increase	3.2 #5	Claim

	project activity. This is alongside training on how to use the ICS, its advantages, as well as how it contributes to climate change and the carbon cycle. The local staff will be annually trained to make sure that the knowledge transferred is up to date and the delivery of the project remains of a high standard.	employment, decent jobs, and entrepreneurship. An total of 24 local staff and personnel were trained during the project lifetime to conduct regular monitoring of the project sites for additional monitoring and maintenance of the ICSs and project activity. This is alongside training in how to use the ICS, its advantages, as well as how it contributes to climate change and the carbon cycle. The local staff will be annually trained to make sure that the knowledge transferred is up to date and the delivery of the project remains of a high standard.		of skill within the project area.			
5)	During this monitoring period, the proportion of time spent cutting, collecting and carrying firewood from trees far away from households was monitored to be now 1.78% of a day for people staying in the project area, compared to 6.25% in the baseline scenario. This decrease from the baseline scenario results in a total time	The project will reduce women and children's drudgery through time savings in reducing time spent on cutting, collecting, and carrying firewood from trees far away from households. Monitored at 6.25% of a day in the baseline scenario as per Tasman Environmental Market's Baseline Survey	5.4	5.4.1 Proportion of time spent on unpaid domestic and care work, by sex, age, and location	Implemented activities to decrease	3.2 #3	Claim

	saved on un-paid work to be 1.07 hours per day.	Report, participating households were now monitored to spend only 1.78% of a day collecting firewood on average. As such, this decrease from the baseline scenario results in a total time saved on unpaid work to be 1.07 hours per day.					
6)	Out of the total 24 local individuals that has been trained and hired for this project, 37.50% of them are female.	Ensure women's full and effective participation and equal opportunities for leadership at all levels of decision-making in the project. During the lifetime of the project activity, a total of 24 local individuals that has been trained and hired for this project, 9 (37.50%) are female. Monitored as the fair and unbiased employment of women within the local team during the lifetime of the project. The project envisages that 50% of its employees will be female.	5.5	Goal 5: Number of women in the project including managerial roles	Implemented activities to increase	3.2 #3	Claim

7)	As of this current monitoring period, a total of 19,206 traditional cookstoves were replaced with fuel efficient ICSs in the Southern Highlands project area.	The project contributes to increasing access to ICS technology with Burn cookstove installations in 19,206 households that have been using traditional three-stone fire, over the project lifetime.	7.1	Goal 7: Number of people with primary reliance on clean fuels and technology	Increase	3.2 #1	Claim
8)	During this current monitoring period, a total of 2 full-time and 22 part-time local staff were hired. Out of the 24 staff, a total of 37.50% are females. A ratio of male to female employees is 5:3 All the staff will be annually trained to make sure that the knowledge transferred is up to date and the delivery of the project remains on high standards.	With the inception of the project activity, jobs will be created firstly for the distribution of ICSs. Personnel will also be trained in how to use the ICS, its advantages, as well as how it contributes to climate change and the carbon cycle. Further to that and aside from the monitoring done during the verification cycle, the project activity will also conduct regular monitoring of the project sites for additional monitoring and maintenance of the ICSs and project activity. In turn creating long term jobs for the local community, without prejudice of sex, age, occupation and persons with disabilities.	8.3	Goal 8: Number of people formally employed, by sector and sex under the project activity.	Implement activities to increase.	3.2 #4	Claim

		During the lifetime of the project activity, a total of 2 full-time and 22 part-time local staff were hired. Out of the 24 staff, a total of 37.50% are females. All the staff will be annually trained to make sure that the knowledge transferred is up to date and the delivery of the project remains of a high standard.					
9)	In this monitoring period, out of the total of 24 staff, 22 personnel were employed part-time. These part-time employees were compensated with an average income of (1,343.33 Kina / month) and is pro-rated where part-time employment is relevant. This value considers age, sex, occupation and persons with disabilities.	Contribute to an increased economic income by providing fair compensation per month for the employees performing the jobs created as part of the project activity initially during deployment and later during project monitoring. Compensation will differ depending on the job scopes assigned to each formal employee employed under the project activity and will be sampled based on this distinction. Individuals employed under the project activity will likely be community members who are currently unemployed or	8.5	8.5.1 Average hourly earnings of employees, by sex, age, occupation, and persons with disabilities	Implement activities to increase.	3.2 #4	Claim

		working in the informal sector with an average income estimated as K439/month (Bonilla, et al., 2020). During the lifetime of the project activity, a total of 24 staff, 22 personnel were employed part-time. These part-time employees were compensated with an average income of (1,343 Kina / month) and is pro-rated where part-time employment is relevant.					
10)	Total emission reduction in the current monitoring period is 13,443 tCO2 .	Contribute to GHG emission reduction through a reduction of ~3.20 tCO₂e per ICS due to the replacement of baseline stoves with Burn ICSs over the crediting period. Over the lifetime of the project activity, a total of 19,206 ICSs were distributed, generating a total of 13,443 tCO₂ eq.	13.0	Tonnes of greenhouse gas emissions avoided or removed	Increase	VCS validation report	SD VISta-labeled VCU
11)	The project has resulted in saving 3.70 tonnes of non-	Contribute an estimated reduction in removal of woody biomass to the tune of 3.70 tonnes of firewood per ICS per	15.2	15.2.1 Progress towards sustainable forest management through the increase of	Implemented activities to increase	4.2 #1	Claim

	renewable biomass per year per ICS. Over this monitoring period, a total savings of 13,446 tonnes of biomass from 19,206 ICSs distribution.	year, from forests surrounding the communities thereby leading to an increase in above-ground biomass in these forests. The above value is based on the difference between the amount fuel used with open-fire cooking ($BC_{b,i,y}$) and the amount of fuel used with the ICS ($BC_{p,j,k,y}$) based on the VM0050 Methodology. Over the lifetime of the project activity, a total savings of 13,446 tonnes of biomass from 19,206 ICS installations.		above-ground biomass in forests				
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2 PROJECT DESIGN

2.1 Project Objectives, Context and Long-term Viability

2.1.1 Summary of Project Sustainable Development Objective(s)

Before the implementation of the project in Papua New Guinea, most of the local people in the island used non-renewable woody biomass for their fuel needs, for cooking with open fire or three-stone fire. The project activity will distribute fuel-efficient ICSs to replace the baseline cookstoves in households. Of which the ICS disseminated through this project will replace open-fired wood-fuelled cooking methods.

The project has been enabling and enhancing households to achieve several Sustainable Development Indicators (SDI):

The project provides ICSs with high thermal efficiency to reduce the consumption of firewood which helps achieve efficient use of an important type of nature resources, non-renewable biomass. This in turn impacts the estimated reduction in removal of woody biomass to the tune of ~3.70 tonnes of firewood per ICS per year, from forests surrounding the communities thereby leading to an increase in above-ground biomass in these forests. This value is estimated from the difference between the amount fuel used with open-fire cooking ($BC_{b,i,y}$) and the amount of fuel used with the ICS ($BC_{p,j,k,y}$) based on the VM0050 Methodology. (SDI15.2.1)

The project activity aims to replace a total of 100,000 traditional cookstoves with fuel efficient ICSs. The efficiency of changing from a traditional cookstove to a more efficient one will result in the consumption of lesser firewood that is usually purchased at high prices. Hence reducing the economic burden of families under the project activity. (SDI Goal 1)

The distribution of ICSs will also introduce and increase access of the local population to ICS technology with Burn cookstove installations in around 100,000 households that have been using traditional open fire cooking, over the project lifetime. (SDI Goal 7)

It will also contribute to the improved health and well-being brought about by reduced levels of fine particulate matter ($PM_{2.5}$) emissions within households below the baseline emission level of 943.30 mg/MJ. (SDI Goal 3)

With the inception of the project activity, jobs will be created firstly for the distribution of the ICSs. Further to that and aside from the monitoring done during the verification cycle, the project activity will also conduct regular monitoring of the project sites for additional monitoring and maintenance of the ICSs and project activity. This creates long term jobs for the local community, without prejudice of sex, age, occupation, and persons with disabilities. TEM envisions that 50% of the project's employees will be female. (SDI Goal 8 and SDI8.5.1)

With the creation of jobs, the project proponent will also carry out training for vocational and relevant skills of local individuals by providing non-formal education and training on issues

related to climate change, with specific skill building in operations and surveying activities related to ICSs distribution and its monitoring under VCS. (SDI Goal 4.1 and SDI Goal 4.2)

All in all, this contributes to an estimated average annual GHG emissions reduction of 3.20 tCO_{2e} per ICS due to the replacement of baseline stoves with Burn Cookstoves over the crediting period. (SDG 13)

2.1.2 Description of the Project Activity

The Fuel-Efficient Cooking in Papua New Guinea project involves distribution of fuel-efficient ICSs in Papua New Guinea (PNG). The ICS disseminated through this project replaces open-fired wood-fuelled cooking methods.

The project is undertaken by TEM PNG Cookstoves Project (“TEM”), during which 100,000 ICSs will be distributed free-of-charge to households in project area. TEM PNG Cookstoves Project Pte Ltd is also dedicated in creating additional value through this project, enhancing the local community’s health, well-being, and understanding and awareness of climate change and sustainable development.

The primary activity under this grouped project is the installation of ICSs.

Improved Cookstoves (G3 Burn rocket stove (also known as Kuniokoa or Ecoa wood cookstove) with adjustable pot skirt):

The primary project activity is the distribution and installation of the G3 Burn rocket stove, utilizing an air-cooled design in order to keep the inside of the combustion chamber hot and the outside cool, increasing durability, performance, and efficiency. Increased efficiency in burning woody fuel would mean less wood being used during the combustion process, encouraging local household to transition away from traditional open fire cooking to cleaner, more efficient cooking solutions.

Training:

In terms of training, TEM PNG Cookstoves Project conducted a two-day course⁸ for its trainers on climate change including the impacts of household pollution on health and well-being, as an introductory background to the efficient cookstove intervention. Training generates positive impacts on community groups by enabling local community members to build alternative skills and build capacity. These interventions are expected to advance socio-economic status, generate greater community capacity for sustainable livelihoods, and increase climate resilience.

During ICS distribution, these trainers assist stakeholders in understanding:

1. How the project activity works and its associated socio-economic benefits

⁸ Training course deck available to VVBs on request. This two-day training is only for new hires, existing hires will undergo a one-day refresher course before being sent for any field work.

2. How to use the ICS
3. End-User Agreement between TEM PNG Cookstoves Project Pte Ltd and End-Users

2.1.3 Implementation Schedule

Date	Milestone(s) in the Project's Development and Implementation
31/01/2023	Consultation with Climate Change and Development Authority (CCDA) on Project Design and Local Stakeholder Consultation
31/01/2023	Consultation with PNG Forestry Authority (PNGFA) on Project Design and Local Stakeholder Consultation
13/03/2023	Local Stakeholder Consultation in Southern Highlands Province
15/03/2023	Local Stakeholder Consultation in Western Highlands Province
17/03/2023	Local Stakeholder Consultation in Jiwaka Province
20/03/2023	Local Stakeholder Consultation in Eastern Highlands Province
30/06/2023	Listing of Project under the SD-VISta
13/07/2023	Listing of Project under the VCS Program
13/08/2023	Undergoing validation (Public comment period closed)
01/01/2024	Estimated Project Start date: Project Activities commence with G3 Burn rocket stove (known as Kuniokoa Cookstove) distribution and installation
14/06/2024	Project Registration Approval under the VCS Program
29/08/2024	Project Registration Approval under the SD-VISta Program
23/09/2024	Project Start Date
30/09/2024	Start Date for Monitoring Period Commissioning of Batch 1
31/10/2024	Start Date for Monitoring Period Commissioning of Batch 2

2.1.4 Project Proponent

Organization Name	TEM PNG Cookstoves Project Pte Ltd
Role in the Project	Project Developer
Contact Person	Dr. Gabriel Machovsky Capuska
Title	Director of International Projects
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2.1.5 Other Entities Involved in the Project

TEM PNG Cookstoves Project Pte. Ltd. is the sole entity involved in the project.

Organization Name	Not Applicable
Role in the Project	Not Applicable
Contact Person	Not Applicable
Title	Not Applicable
Address	Not Applicable
Telephone	Not Applicable
Email	Not Applicable

2.1.6 Project Location

The GPA will distribute ICSs to communities within the project boundary of PNG (Latitude: 6.3150° S Longitude: 143.9555° E). The first project instance monitored in this monitoring report is within the Southern Highlands Province (see *Figure 1 & Table 2*).

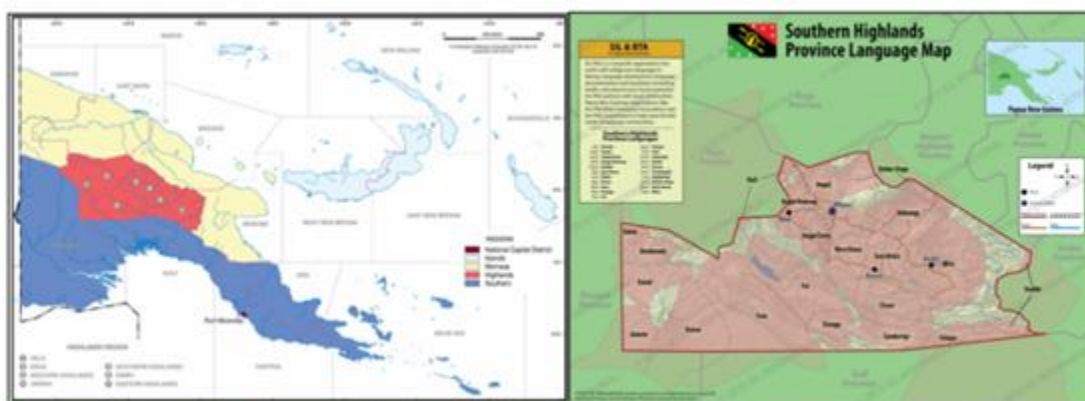


Figure 1: Map of Papua New Guinea including the Highlands Region (left) and map of Southern Highlands Province (right). Maps extracted from www.pnglanguages.sil.org

Table 2: The Geographic coordinates of the two provinces involved as the first project instances of the group activity in Papua New Guinea

S/N	Location	Latitude	Longitude
1.	Southern Highlands Province	6.4179° S	143.5636° E

2.1.7 Project Description Deviations

According to the Project description, monitoring of PM_{2.5} was to be conducted through measurements taken in the household during sampling within the households' kitchen area during a cooking activity. Reliable PM_{2.5} measurements in household environments are a technical challenge due to large spatial variabilities across households, differences in cooking practices, inconsistent data from low-cost sensors, difficulty in isolating indoor sources from outdoor infiltration, the need for long-term monitoring to capture seasonal variations, and the complexity of accurately capturing personal exposure due to movement within the house⁹. All of these can significantly impact the accuracy of PM_{2.5} measurements within a household environment.

In the project description deviation, it was decided to use the results from the extensive standardised testing (using the ISO Protocol 19867-1:2018) available in the Clean Cooking Alliance database¹⁰ for the specific model of the project stove, comparing the with the same parameters for open fire cooking. The comparison of supporting evidence from the database catalogue on clean cookstoves demonstrated a reduction in PM_{2.5} in households to be by 81.24% from 943.40 mg/MJ¹¹ to 177 mg/MJ¹² for newly distributed ICSs. Hence, calculation on the decrease in PM_{2.5} is possible based on this literature, project specific parameters and the number of ICSs distributed. A detailed calculation is available upon request.

Secondly, it is understood that the thermal efficiency of an ICS is expected to decrease year on year. This decrease will be tested annually through the Water Boiling Test stated in VM0050 v 1.0 conducted by the ICS manufacturer and will be used in the detailed calculation to ensure an accurate calculation of the decrease in PM_{2.5}. However, for the first monitoring period, all ICSs are expected to be working at maximum efficiency as per the manufacture's standards as the monitoring duration is less than 12 months.

Thirdly, according to the Project Description, the estimated reduction in removal of woody biomass was 3.81 tonnes of firewood per ICS per year. With the release of the new methodology, VM0050 v1.0, the estimated reduction in removal of woody biomass is obtained by taking the difference between the amount fuel used with open-fire cooking with the amount of fuel used with the ICS. This gives us an updated value of 3.70 tonnes of firewood per ICS per year saved which is used during calculations.

Lastly, in order to qualitatively account for and supplement evidence for the decrease in PM_{2.5}, during the Ground Monitoring Surveys, questions on the type of smoke emitted during cooking:

⁹ Assessing household fine particulate matter (PM_{2.5}) through measurement and modeling in the Bangladesh cook stove pregnancy cohort study (CSPCS), M.M. Rahman et. al, 2023, <https://www.sciencedirect.com/science/article/abs/pii/S0269749123015701>

¹⁰ Clean Cooking Alliance Database: <http://catalog.cleancookstoves.org/>

¹¹ http://catalog.cleancookstoves.org/stove_details.html?stove_id=stove_A8WGMJX

¹² http://catalog.cleancookstoves.org/stove_details.html?stove_id=stove_AW0UJ

(1) Thin and Light, (2) Medium and Grey and (3) Thick and Dark was surveyed to households. This was compared to the baseline where only Thick and Dark smoke was observed, to determine if there was a decrease in intensity of smoke produced.

Results from this qualitative survey conducted for the monitoring period showed that 100% of end-users observed a decrease in the smoke intensity compared to the baseline scenario.

2.1.8 Threats to the Project

Table 3: Nature-induced threats

Threat:	Continued deforestation and degradation make firewood an untenable source of biomass.
Solution:	As much as the efficiency of the cookstoves coupled with the imminent scarcity of the wood fuel would support the use of G3 Burn rocket stove (also known as Kuniokoa Cookstoves) as proposed by TEM PNG Cookstoves Project Pte Ltd for this project, there is the potential that as the population continues to expand and there may be a point when firewood becomes an untenable source. In response to this, TEM PNG Cookstoves Project Pte Ltd in country team will train end users to maximise the use of small twigs and branches, crop residues near the local communities they live.

Table 4: Human-induced threats

Threat:	End users' lack of maintenance of ICS resulting in either malfunctioning of the ICS as designed or lack of use.
Solution:	TEM PNG Cookstoves Project Pte Ltd will conduct routine visits to end-user households at least twice a year in each main geographic areas that currently use the new G3 Burn rocket stove (also known as Kuniokoa), to identify potential defects in the stove's construction, maintenance, and sub-optimal use. Additionally, the end-users also have access to a grievance redress mechanism (Section 2.2.7) to report damaged cookstoves to the project staff. To reduce this proposed treat, TEM PNG Cookstoves Project Pte Ltd's in-country team will hire, train, and supervise users that will involve visiting the end-user's houses every 3-months to monitor the use of the new ICS.
Threat:	Failure of Behaviour Change resulting in high levels of non-adoption

Solution:	TEM PNG Cookstoves Project Pte Ltd will continuously research the impacts of behaviour change for successful adoption. Study outcomes, whether formal or informal, will be included in future training and community engagements to continue shaping knowledge, and enhancing positive peer-to-peer influence. Planned community incentivisation mechanisms to reduce the use of inefficient baseline devices and practices are explained below: - Field assistants known as ‘Cookstove Ambassadors’ will actively interact with end-users from their allocated jurisdictions to encourage the use of ICSs and continuously foster awareness on the benefits of using ICS over open fire cooking methods. The increase in adoption rates will be monitored through surveys inquiring about the rate of stove stacking within the jurisdiction. - Monitored results will be tabulated quarterly, and the village with the most progress adopting from open fire cooking methods will be awarded with a ‘Cook-Off Event’. - The Cook-Off Event is an engagement strategy in the form of a celebration within the village where TEM will conduct activities highlighting the ICSs’ benefits such as the reduction of firewood consumption and the reduction in smoke generated during cooking. - Cookstove Ambassadors will also present successful stories through knowledge sharing sessions with the event’s participants. Additionally, TEM will work with Cookstove Ambassadors seeking successful testimonials from end-users benefiting from clean cooking practices through the use of ICSs that can be converted into short form videos to capture broader social media audiences. Non-adopters, where identified, will also be engaged to understand the potential barriers preventing them to adopt the use of the ICS. These outcomes and conclusions will be useful to improve future training events with the goal of avoiding miss-information.
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Threat:	ICS Projects and distributions are not shown to have an increased impact on household uptake.
Solution:	Time savings and increased health benefits are co-benefits from the adoption of ICSs will be incentives to increase their uptake. In response to this, TEM PNG Cookstoves Project Pte Ltd will conduct trainings highlighting

	how time saving, health benefits, previous users' positive experiences are important factors for large scale adoption.
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2.1.9 Benefit Permanence

To ensure the project's long-term viability, the permanence of the individual ICS and other program benefits, TEM PNG Cookstoves Project Pte Ltd will take the following long-term approach:

- 1) Periodical monitoring of the end users to ensure the regular use of ICSs. To guarantee the success of this process our local staff will conduct visits to the communities multiple times per year. These teams will be provided with transportation and tablets that meet the physical mobility and remote connection requirements of their roles.
- 2) Conduct training sessions for the local communities to increase the awareness levels of these communities regarding the long-term environmental, health, and resiliency benefits aligned with the adoption of ICSs. The understanding of these holistic benefits to the stakeholders is valuable for ensuring long-term sustenance and maintenance of project activities and achieving the intended outcomes.
- 3) TEM PNG Cookstoves Project Pte Ltd and the Western Pacific University (WPU) signed a Memorandum of Understanding (MoU) (see file titled: MoU_TEM and WPU_PNG Cookstoves project signed in the PD) to collaborate on the development of supporting environmental and sustainability activities directed to educate and train local students in topics related to sustainability, climate change impacts and mitigation strategies. This will provide the opportunity to educate local youth on the benefits of clean energy and clean cooking practices and to better understand the extent of the PNG ICS project as a mitigation strategy for climate change.

Selected local students from WPU will gain working experience (i.e. casual, part-time or full-time employment) in different stages of project implementation in a range of transferable skills (e.g., technology IT support, administration, project management, community engagement, data collection and curation).

This partnership also contemplates the involvement of local university students and end-users designated as ICS ambassadors in the potential development of outreach activities with primary and secondary schools in the area to foster awareness of the health, environmental and economic benefits of using ICS for cooking.

As stated in the MoU, the main goal is to involve the local communities in the proposed project activities, creating a sense of ownership amongst the people, and inspiring behavioural change from a young age through formal education.

These efforts will ensure that over the duration of the project the end users will have a positive behavioural change through the improved awareness, education and first-hand experience of the health and economic benefits over traditional open fired cooking, which will in turn lead to the continued use of ICSs even after the period of project activity. Evidence for benefit permanence stated above is further discussed in section 3.1.

2.2 Stakeholder Engagement

2.2.1 Stakeholder Consultation and Adaptive Management

During this monitoring period, stakeholder consultations presented are focused on the provinces where distribution of ICSs took place.

Date of stakeholder consultation	Southern Highlands Province: 13 March 2023
Stakeholder engagement process	Prior to the LSCs, TEM utilized several media outlets to advertise the event to the provinces. These efforts were fulfilled using: • Newspaper Advertisements • Letters of Notification to Provincial Governments • Radio Broadcasts • Invitation Letters to Churches • Social Media Posting • Webinar • Email to PNG Australia Alumni Association (PNGAAA) members All advertising outlets were run in a staggered format with the aim of keeping stakeholders informed and aware of the LSCs that would be happening within their provinces. For the exact dates of advertisements being published, please refer to the PD Appendix II: LSCs Outreach. The stakeholders engaged in this project were people or groups of people that were directly or indirectly affected by the project as well as those who were interested in the project and or had the ability to influence its outcome either positively or negatively. They include: • Representatives from the provincial administration • Representatives from Women Groups • Representatives from Churches • Ward councilors • Heads of households both male and female

- Key informants such as teachers, community health workers etc.
- Improved Cookstove (ICS) users (SHP only)
- Youth leaders and
- Educated elites (lecturers, nursing officers, Health Extension Officers, Police personals etc.)

Southern Highlands Province LSC

A total of 159 people attended this LSC on the 13th of March 2023 held at Kumin Catholic Mission in Mendi, SHP. Including those who commuted into Mendi from various districts, there were public servants such as nursing officers, police, teachers and church elders and pastors present during the consultation.



Figure 2: A demonstration by Ms. Magret Yangpin and Mr. Richard Manda – SHP Field Assistants.

The LSC was facilitated by Mr. Michael Kisombo, the project Manager with Simi Oa Consultancy Services who was engaged by TEM, did a power point presentation after welcoming and acknowledging all the stakeholders that were present. The PowerPoint slides were prepared in English, and explained in tok pidgin, the common language where most could understand well as those participated came from different local dialects.

Attendees were briefed on the purpose of the meeting, and informed of the:

- Carbon cycle and the important role it played within their forests.
- Findings of the Baseline Study conducted in the Highlands region including SHP and WHP (file titled:

	TEM 2025_Cookstoves Highlands region report_v3.pdf) - The Verra registration process - The role of carbon finance in the project activity - Indicative timeline and implementation plan of the project Two field assistants as seen in Figure 2 were then invited to demonstrate and to explain how the ICS is used using the sample that was exhibited during the LSC. That was further reinforced by one of the users who was also invited to give her experience of how she is currently using the ICS and explained the benefits her family is experiencing. After the presentation, stakeholders were given the opportunity to ask questions and provide their reflections. For the summary of questions asked and subsequent answers provided by TEM, please refer to Appendix III: Stakeholder Feedback. After the discussions, there were about 29 feedback forms that were received through the feedback forms. These feedback forms may be made available to validators upon request.
Consultation outcome	There was no negative feedback received during the LSC. Most questions aimed to seek clarity on how a more efficient ICS could reduce carbon in the atmosphere, when the project could begin, and if more ICSs could be distributed. For a complete list of questions and feedback, please refer to the PD Appendix III: Stakeholder Feedback. For more information, a Local Stakeholder Consultation Report is available upon request of the validator.
Ongoing communication	During the Local Stakeholder Consultations, it was discussed how stakeholders may keep in contact with TEM PNG Cookstoves Project Pte Ltd's in country project manager, Michael Kisombo. During which, the following solutions were offered: 1. Reaching out to the in-country Team during the twice a year visits to maintain the ICSs. 2. Reaching out to TEM PNG Cookstoves Project Pte Ltd Pacific through the project's dedicated email. 3. Reaching out to TEM PNG Cookstoves Project Pte Ltd through the project's website. Stakeholder consultations conducted during the monitoring period, please refer to Appendix I: Stakeholder Consultations

	The following website of the project activity was available to the stakeholders: https://www.tasmanenvironmental.com.au/pngcookstoves/ The following email addresses will be also used for inquiry or comment on the project: pngcookstoves@tem.com.au TEM PNG Cookstoves Project Pte Ltd in-country Project Manager: Michael Kisombo, +675 7262 3864, michael.kisombo@tem.com.au.
Stakeholder input	Please refer to the PD Appendix III: Stakeholder Feedback.

For more information on stakeholder consultations conducted during the monitoring period, please refer to Appendix I: Stakeholder Consultations.

Overall, TEM engaged with ICS end-users during visits to register the households and to give out the ICSs. In total, TEM has registered and distributed a total of 19,206 ICS cookstoves during this monitoring period. In addition, monitoring surveys and training of new staff was also conducted. In addition, monitoring surveys and training of new staff were also conducted where the results are further described section 3.1.

2.2.2 Anti-Discrimination

TEM PNG Cookstoves Project Pte Ltd will distribute ICSs to local people who uses traditional low efficiency cooking solutions without distinction in genders, races, religions, educational backgrounds, or any other aspects. TEM PNG Cookstoves Project Pte Ltd is dedicated to the elimination of discrimination, and as jobs are created during the implementation of this project, the company will ensure that women, persons with disabilities, and minority will have equal chance to get these jobs.

All TEM Employees adhere to the TEM Employee Handbook which instils the importance of preventing Harassment (Section 13) and Equal Opportunities and Anti-Discrimination (Section 14). This handbook informs and ensures that all employed under TEM ensures that any form of harassment or bias in project operations is prevented by clearly laying out the definitions of harassment and discrimination, and its consequences.

These policies were enforced throughout the monitoring period.

2.2.3 Worker Training

The implementation of the project needs plenty of local people to participate in production, distribution or use steps. They will get relevant skills and sustainable development and global citizenship education through training by project proponent. Besides technical skills, the workers will be trained about the management structure, regulations, and worker rights.

TEM PNG Cookstoves Project Pte Ltd organised a two-day training course¹³ between 08 July 2024 and 11 July 2024 for the in-country team to learn about monitoring techniques including survey methods, data record and analysis. The monitoring plan was carried out by qualified personnel trained for quality assurance and quality control.

For inclusivity, training such as the technical skills and education on regulations were provided for marginalized and vulnerable group as 37.50% of the individuals trained were female.

To ensure long-term skill retention, a two-day training course refresher prior to monitoring was conducted from 15-Jan-2025 to 17-Jan-2025 for the field team. To ensure continuity in this, a refresher is always to be conducted prior to any monitoring.

More details on these trainings are available in Appendix 1.

2.2.4 Equal Work Opportunities

TEM PNG Cookstoves Project Pte Ltd is dedicated to make sure that all stakeholders, no matter their genders, races, religions, educational backgrounds, or any other aspects, will be given an equal opportunity to fill all work positions. Recruitment advertising for the local villages will be given before the hiring of the full-time and part-time staff for the project. The recruitment advertisement will be provided in Pidgin English for them to understand.

These policies were enforced throughout the monitoring period and hiring was done fairly in partnership with the recruitment agency, Vanguard, which was in charge of recruiting, interviewing and contracting full-time employees according to TEM's Job Description and Equal Opportunities & Anti-discrimination Policy found in the TEM Employee Handbook, Section 14.

During the monitoring period, it was noted that 37.50% of the individuals hired were female.

2.2.5 Workers' Rights

TEM PNG Cookstoves Project Pte Ltd has a labour contract with each in-country personnel following PNG labour legislation. Before the workers sign it, the contracts are explained explicitly to make sure that they could understand their rights and obligations. The contract terms conform with, and uphold the principles and rights of, work addressed in the Core Labour Conventions of the International Labour Organization (ILO), as well as comply with national and local labour laws within Papua New Guinea.

During the monitoring period, each worker is placed in teams where there is a hierarchy for reporting. During the hiring process, it is known to the workers (regardless of position) that reporting of any labour rights violations or grievances is possible through reporting it to a superior.

¹³ Training course deck available to VVBs on request. This two-day training is only for new hires, existing hires will undergo a one-day refresher course before being sent for any field work.

These policies were enforced throughout the monitoring period.

2.2.6 Occupational Safety Assessment

To mitigate potential occupational safety hazards, TEM PNG Cookstoves Project Pte Ltd makes the effort to contract workers who have lived in the community for several years, thus assisting in mitigating the occupational hazards through community familiarity, language fluency, and being native to the culture. This understanding of traditional values, respect, and working environment in the communities TEM PNG Cookstoves Project Pte Ltd serves to support reduction of safety hazards.

Risks are assessed and mitigated where possible and workers, stakeholders, and others directly involved with the projects are made aware of the potential risks and opportunities to mitigate where possible, yet it should be noted that employment or use of the ICS does not increase the safety risks that are already present in the landscape of the country.

TEM ensures safe and appropriate working conditions for all its employees through the development and consultation of its Project Risk Assessment. Within this assessment sheet, risks related to transportation and security are identified for stakeholders including TEM staff, the consequences of risks to occur and their likelihood of it happening is stated via an impact score. Then, mitigation actions for these risks are included to bring the residual risk down to acceptable levels.

Additionally, all personnel who travel into the project boundary must adhere to the tailored security travel plans that are developed with clear emergency protocols to guarantee their safety. The travel plan includes details for Remote Check-Ins with a designated Check-In person, full travel itinerary, emergency contact details, etc., and must be pre-approved by TEM's Director of International Projects before they are circulated to all parties participating in the trip.

2.2.7 Feedback and Grievance Redress Procedure

No grievances were reported during the monitoring period.

The details on procedure for feedback and grievance have been made publicly available to all stakeholders on the project website of TEM PNG Cookstoves Project Pte Ltd <https://www.tasmanenvironmental.com.au/pngcookstoves/>

Additionally, a Grievance Redress Mechanism (GRM) Standard Operating Procedure (SOP) has been prepared by TEM PNG Cookstoves Project Pte Ltd and aims to present a set of guidelines for designing and implementing different mechanisms that can address a variety of grievances arising when installing the ICSs in PNG during the crediting period of the project activity. The guidelines are prompt and effective and are timely to inform the complainants of the actions taken to resolve issues raised. The channels for reporting the grievances and the process of categorising common grievances are provided below.

Table 5: Channels for Grievance Reporting

Grievances Reporting Channel	Grievance Reporting Mechanism
Physical walking in	Complaints to TEM PNG Office and/or its staff
Social media and online	Complaints made online to TEM PNG Office, and/or in country Project manager
Phone call	Phone calls to Field Assistance, and/or Project manager
Verbal through community representatives	Complaints through community representatives who then channel through the organizational structure

Table 6: Process of organizing common grievance

Type of Grievance	Scope	How will be facilitated	Who will handle	Timeline
Unfair distribution of ICS	Cookstoves unfairly distributed by project staff	Gather information, investigate and refer to project lead	In country Project manager	7 -14 days
Burglary, sold or destroyed by natural disaster	Cookstoves were stolen, destroyed, or sold	Gather information, investigate, and refer to project lead	Project manager and project lead	14 – 21 days
Household missing out on the distribution	Cookstoves missed targeted households	Investigate and refer to project lead for possible re-issue	As above	5 – 7 days
Damaged cookstoves	Cookstoves issued were damaged or have technical issues	Investigate and refer to project lead for possible replacement	As above	5 -7 days
Complaints raised by project staff	Unfair decisions related to pay or disciplinary actions	Gather information and refer to project lead	Project manager	5 – 7 days

With reference to the GRM, all complaints and grievances will be registered and monitored by the in-country Project Manager supported by the field assistants. These complaints and grievances will be resolved using traditional and customary ways.

This document will be socialized and publicized to communities throughout the project location in their local language, Pidgin English.

During the monitoring period, no formal complaints were received or logged from the end-users in the form of physical walk ins, social media and online postings or phone calls, and verbally through community representatives.

Verbal feedback has been very positive, while one feedback was made in the presence of the validator during the site visit regarding soot from open fire cooking being able to maintain the waterproofing of the end-user's dwelling. However, as the outcome of using the project ICS in lieu of open fire cooking cannot be scientifically proven, this feedback has been recorded as informal feedback.

2.2.8 Stakeholder Access to Project Documentation

TEM PNG Cookstoves Project Pte Ltd will inform the beneficiaries and their families that they are participating in a project that distributes the G3 Burn rocket stove (also known as Kuniokoa cookstove) upon the household's receipt of the ICS. TEM PNG Cookstoves Project Pte Ltd will also inform them of the need to track their successes and monitor key data, such as money spent on wood fuel and/or time spent collecting wood fuel, among other indicators. ICS end-users will also be made aware that to ensure the integrity of the project and programs, auditors, and stove champions, visit the families up to 2 times after implementation.

As for project documents, the full project documentation will be uploaded on VERRA website, and the same ones will be publicized on the TEM PNG Cookstoves Project Pte Ltd website during the crediting period of the project as the following linkage:

<https://www.tasmanenvironmental.com.au/pngcookstoves/>

2.2.9 Information to Stakeholders on Assessment Process

To ensure all the stakeholders are aware of any assessments, including the site visits of assessors, TEM PNG Cookstoves Project Pte Ltd and its in-country team will inform them in advance (at least 7 days) through the website linkage, phone, email and the following communication channels that are effective in the community.

1. Community announcements through local newspapers and radio *toksaves*

Local newspapers: 'The National and Post courier', and radio shows like *Radio nek blong eagle* (Radio Western Highlands Province) have been used to inform about stakeholder meetings in the past and will be continued. These are paid advertisements those run as community announcements prior to meetings.

2. Facebook Forums and WhatsApp groups

There are several Facebook sites and WhatsApp groups created to discuss developmental issues in each province. For instance, SHP Development forum, EHP Development forum, WHP development forum, among others. This has been utilised for previous stakeholder engagement program details including venues for community engagement, time and date and so forth using both English and tok pidgin (national language).

3. Community meeting platforms and word of mouth

Church gatherings are significant places to inform stakeholders about the project. Community announcements will be prepared in Tok Pidgin and issued to Parish Priests, Pastors and Church elders who make the announcements during their church gatherings. As a significant population are illiterate, phone communications to village elders, ward councillors and project field assistants to spread the information are also an effective method to disseminate information.

The meetings of stakeholder with assessors will be facilitated at a venue informed in advance through the above channels and in the presence of an adequate interpreter (if required due to language restrictions) who would be a community member but not associated to the project.

2.3 Project Management

2.3.1 Avoidance of Corruption

TEM PNG Cookstoves Project Pte Ltd staff worldwide follows strict and robust rules and regulations regarding the avoidance of corruption, code of conduct, and business ethics.

All TEM employees adhere to the Anti-Bribery and Corruption Policy Laws to protect both themselves and the reputation of the company. Under this policy, TEM employees are not permitted to give, offer, promise, accept, request, or authorise a bribe, whether directly or indirectly or abuse entrusted power for private gain.

Any person or organization which violates these anti-corruption rules and regulations will immediately stop their work engagement with TEM PNG Cookstoves Project team.

2.3.2 Recognition of Property Rights

All ICSs included in this project activity were distributed to individual households free of charge. Before receiving their ICS, end-users will sign a carbon transfer agreement (i.e., End-User Agreement) with TEM PNG Cookstoves Project Pte Ltd, clearly stating that the ICS will belong to them, whereas the carbon credits generated from their use will belong to TEM PNG Cookstoves Project Pte Ltd.

2.3.3 Free, Prior and Informed Consent

The project is voluntarily implemented by TEM PNG Cookstoves Project Pte Ltd, and end-users are free to choose whether they will be involved in the project or not. Free, prior, and informed consent takes place before distribution through signing of the End User Agreement. In the agreement, end-users acknowledge that they have received the ICS, and they will transfer the rights of the carbon credits generated from the project to TEM PNG Cookstoves Project Pte Ltd.

2.3.4 Restitution and/or Compensation for Affected Resources

The project activity involves distribution of ICSs to individual households only and it will not affect any resources.

2.3.5 Property Rights Removal/Relocation of Property Rights Holders

The project activity involves distribution of ICSs to individual households only and it will not lead to any removal of property rights or relocation of property rights holders.

2.3.6 Identification of Illegal Activities

Theft and corruption may be identified during the implementation of the project. TEM PNG Cookstoves Project Pte Ltd and its local partner will avoid any corruption as per Section 2.3.1 of this document. End-users will be reminded to keep their ICSs in safe places to reduce theft when they receive the ICSs.

2.3.7 Ongoing Conflicts or Disputes

Where applicable, describe activities and/or processes implemented to resolve conflicts or disputes.

The project activity involves distribution of ICSs to individual households only. There is no ongoing or unresolved conflicts or disputes over rights to lands, territories and resources and any disputes that were resolved during the last twenty years.

2.3.8 National and Local Laws and Regulations

The project aligns with several International and Papua New Guinean's laws, regulations and conventions. TEM PNG Cookstoves Project Pte Ltd, as the project developer, and its in-country team will follow and obey all these relevant laws, regulations and conventions.

These are stated below:

International Laws, Regulations and Conventions

- **United Nations Framework Convention on Climate Change (UNFCCC,1992)**

The aim of the UNFCCC is to achieve the "stabilization of greenhouse gas concentrations in the atmosphere at a level that would prevent dangerous anthropogenic interference with the climate system".

The project has been implemented to reduce Green House Gas (GHG) emissions in an effort to combat climate change, and therefore the Project is compliant with the UNFCCC.

- **United Nations Declaration on the Rights of Indigenous Peoples Rights (2007)**

The Declaration delineates the individual and collective rights of indigenous peoples, their land, territories, resources, culture, identity, language, employment, health, education, and other issues. It also "emphasises the rights of indigenous peoples to maintain and strengthen their own institutions, cultures and traditions, and to pursue their development in keeping with their own needs and aspirations". It "promotes their full and effective participation in all matters that concern them and their right to remain distinct and to pursue their own visions of economic and social development".

The project is aligned with this United Nations Declaration by providing the local (indigenous) communities with the opportunity to engage, at their own free will, in economic development via

carbon finance, protection of their lands, territories and forest resources while also enhancing opportunities for other livelihoods, education, employment and healthcare. Furthermore, the project aims to improve the general way of life for current and future generations, including education on traditional ways of *life* and enhancing community by way of communication, meeting places and knowledge sharing.

- **Stockholm Declaration of the United Nations Conference in Human Environment (1972)**

The Stockholm Declaration states that it is each person's right to freedom, equality, adequate conditions of life; responsibility to protect the environment for present and future generations; to safeguard natural ecosystems and ensure the capacity of the earth to produce renewable resources is maintained, restored, and improved.

The project provides a mechanism whereby the local community and the Papua New Guinean Government can protect the ecosystems and environment of the Project Area, and lead to an improvement in the socio-economic conditions of the community. In addition, the renewable resources of the area will be maintained and possibly improved by the implementation of the project. The socio-economic well-being of local communities will also be improved through income and community development programs.

- **Rio Declaration on Environment and Development (1992)**

The 'Rio' Declaration aims for people to have a healthy and productive life in harmony with nature. It aims to ensure that development activities within national jurisdiction do not cause damage in areas beyond the national boundary. It also aims to eradicate poverty and encourage sustainable development and environmental protection.

The project aims to avoid GHG emissions within Papua New Guinea, which would have had an adverse impact on the global climate extending beyond the country's boundaries. The project also encourages an increase in sustainable livelihoods and education to combat poverty in the project area through sustainable development and environmental protection practices.

- **Paris Agreement (2015)**

The Paris Agreement's goal is to keep the rise in mean global temperature to below 2 °C above pre-industrial levels and preferably limit the increase to 1.5 °C to substantially reduce the effects of climate change, and for emissions to be reduced as soon as possible and reach net zero by the middle of the 21st century.

In alignment with this agreement, the project activities reduce GHG by avoiding carbon emissions.

Papua New Guinea's Laws, Regulations and Conventions

- **Papua New Guinea's Constitution (1975)**

The Project is compliant with Papua New Guinea's Constitution, specifically the fourth goal of the Constitution which is "to ensure that the forest resources of the country are used and replenished for the collective benefit of all Papua New Guineans, now and for future generations".

The project aims to reduce the dependence of the people on firewood and in turn avoid deforestation.

- **Climate Change Management Act 2015, Amended (2022), and proposed Carbon Market draft regulation (2023)**

The Act established the Climate Change Development Authority (CCDA) and provided a framework for the development and implementation of measures in Papua New Guinea to combat climate change impacts in accordance with UNFCCC and other international agreements and programs. The project is aligned with the CCDA policies and the proposed carbon market draft regulations.

- **Forestry Act (1991)**

The Act is aimed at managing, developing and protecting Papua New Guinea's forest resources; maximising participation in and development of forest resources as renewable assets; utilising the forest resource to achieve economic growth, employment and industry; and encouraging scientific study and research to contribute to sound ecological balance.

The project assists Papua New Guinea in achieving its goals under the Act to protect and use forest resources through the reduction of firewood consumption and the development of carbon credits; while achieving economic growth through direct employment opportunities, education, and skills development in the project area.

- **Environmental Act (2000)**

The Environmental Act outlines the protection of the environment in accordance with the Fourth National Goal of the Constitution; promotes sustainable development of the environment and economy; states that the social and physical well-being of the people should be supported by the safeguarding the life-supporting capacity of the air, water, soil, and ecosystems for present and future generations, and avoiding, mitigating any adverse effects of activities on the environment.

The project complies with requirements of the Act to promote sustainable development of the environment (protection) and economy (carbon credits, improved cooking) while supporting the social (training, community health) and physical wellbeing (healthcare) of the local communities. The project also reduces the deforestation through reduced demand of firewood.

- **Investment Promotion Act (1992)**

The Papua New Guinea Investment Promotion Authority (IPA) was established by the Investment Promotion Act 1992 to promote and facilitate investment within Papua New Guinea and abroad. The act regulates registration of companies/businesses and the certification of foreign enterprise. The Act states that investors are required to obtain licences for their area of business interest from the relevant Government agencies in the country.

The project is a mechanism for foreign investment in Papua New Guinea's sustainable forests and community development. The project developer has obtained the appropriate licences to operate in their area of businesses.

- **Industrial Relations Act (1962)**

This Act details the settlement of industrial disputes. It is an Act relating to development of industrial relations and the prevention and settlement of industrial disputes.

The project is aligned with this legislation through its own Grievance Redress processes including a formal dispute resolution strategy.

- **Industrial Health Safety and Welfare Act (1962)**

This Act relates to industrial safety, health and welfare, and related purposes. It ensures that measurements have been undertaken to ensure the health, safety, and welfare of employees in the industrial sector.

The project will take all necessary steps to ensure the working health and safety procedures for all the staff involved in the proposed activities.

- **Employment Act No. 54 of 1978 (Consolidated up to 31 March 2001)**

This Act relates to the employment of certain persons and lays out conditions of employment.

The project has employment policies and procedures that comply with the Act.

- **Child Welfare Act (1961)**

The welfare of children and related purposes are covered under this Act, it covers the protection and promotion of the rights and well-being of a child. Project objectives, including activities aimed at improving health standards and improving educational opportunities, are consistent with the Child Welfare Act.

The Project is committed to recognising the rights and welfare of children within the project area.

- **Discriminatory Practices Act (1963)**

This act is to prohibit certain discriminatory practices. 'Discriminatory practice' in this Act means discrimination either of an adverse or of a preferential kind, practiced by a person or group of persons, against or in favour of another person or group of persons for reasons only of colour, race or ethnic, tribal, or national origin.

The project promotes equal opportunities in the workplace based on personal merit rather than gender, colour, race or ethnic, tribal, or national origin.

- **Workers Compensation Act (1978)**

This Act provides for compensation to workers and their dependants in respect of injuries suffered by workers arising out of or in the course of their employment, and for related purposes.

The project is aligned with this act regarding employment policies and procedures related to workers compensation.

- **Papua New Guinea Vision (2050)**

This document maps out the future direction for Papua New Guinea's socio-economic development. It is underpinned by seven 'Strategic Focus Areas', which are referred to as pillars:

- Human Capital Development, Gender, Youth and People Empowerment;
- Wealth Creation;
- Institutional Development and Service Delivery;
- Security and International Relations;
- Environmental Sustainability and Climate Change;
- Spiritual, Cultural and Community Development; and
- Strategic Planning, Integration and Control

In alignment with Papua New Guinea 2050 Vision, the project will develop environmental and sustainability practices and social development and empowerment to the local communities.

2.4 Grouped Project

This project is a grouped project activity.

New project activities are assessed and evaluated to determine if they provide significant individual impacts (funding of education), household (cookstove), or community level (training on climate change).

For the inclusion of new project activity instances, the project proponent shall ensure that it meets the eligibility criteria for Grouped Projects per the SD VISta Standard as shown in Table 7.

Table 7: Demonstration of compliance towards the eligibility criteria for Grouped Projects

S/N	Criterion	Compliance	Demonstration of Compliance of present instances
1	Adopt and implement the project activities in the same manner as specified in the project description.	New project activity instances will be implemented in the same manner as described in the project description, extending benefits, and reinforcing ICS adoption.	All ICSs distributed in the current monitoring period are G3 Burn Rocket Stoves. They are distributed within the project boundary to maximize adoption rates and awareness of the benefits as mentioned in the project description.
2	Where appropriate, meet the applicability conditions of the SD VISta asset	The project activity, as well as new project instances will apply the SD VISta asset methodology and adhere	The project activity does not apply any SD VISta asset methodology,

	methodology applied to a project.	to the criteria set out in the methodology.	hence, the criteria is not appropriate to this project activity
3	Are subject to the same scenarios at project start with respect to stakeholders' well-being as determined for initial project instance(s), where (per Section 2.1.5.2 above) the project must meet the criteria of Section 3.1 above.	New project activity instances will have the same baseline scenario, which was available during the start of the project, i.e., usage of open fire cooking methods in the households. Also, the instances will be designed to maximize the intended impact and preserve well-being, with monitoring and ongoing stakeholder consultations to ensure impact.	Each household that received an ICS has been using the open fire cooking method in the baseline. Provision of spot audits and surveys enabled the assessment of type and depth of impacts.
4	Are subject to the same scenarios at project start with respect to natural capital and ecosystem services as determined for initial project instance(s) where (per Section 2.1.5.2 above) projects must meet the criteria of Section 3.2 above.	New project instances will have the same scenario as determined in the baseline scenario. All the new instances will also have households relying on wood fuels for meeting their daily cooking energy demands.	All households that received an ICS, revealed using wood fuel for cooking during the monitoring surveys.
5	Are subject to the same processes for stakeholder engagement described in the project description.	The new project activity instances will be detailed during continuous stakeholder consultations, refresher trainings for Implementing Partners and feedback consider in secondary project activity design to maximize the intended impact and preserve wellbeing.	The project proponent has ensured that there have been continuous stakeholder consultations, refresher trainings as well as surveys to maximize intended impacts.
6	Are subject to the same processes for respect for rights to lands, territories and resources including free, prior and informed consent – described in Section 2.4 above.	The outlined Project Management (Section 2.3) will apply to each additional project activity.	Project management for the current monitoring period complies with the structure described in Section 2.3 of Project SD VISta Design document.

7	Have similar monitoring elements to those set out in the project description.	Additional projects will have equivalent monitoring plans and procedures as described in Section 3.3.	The project has implemented the monitoring plan as detailed in section 3.3 and 4.3 of SD VISta DD.
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3 BENEFITS FOR PEOPLE AND PROSPERITY

3.1 Impacts on Stakeholders

Impact #1	Provide access to ICS
Type of Impact	Positive, actual, direct
Affected Stakeholder Group(s)	ICS end-users
Resulting Change in Well-being	Describe characteristics and magnitude of impact on each stakeholder group Baseline: The project activity aims to replace 100,000 traditional cookstoves with fuel efficient ICSs. Upon gaining access to the ICS, inadequate cooking will be gradually reduced. As will the burden of firewood collection, the time to prepare food and the cost to buy firewood, hence improving health and well-being. Monitoring Period: During this monitoring period, a total of 19,206 traditional cookstoves were replaced with fuel efficient ICSs.
Impact #2	Improved Health Status
Type of Impact	Positive, predicted, direct
Affected Stakeholder Group(s)	ICS end-users

Resulting Change in Well-being	Describe characteristics and magnitude of impact on each stakeholder group Baseline: Prior to the project activity, high levels of smoke in the cooking place were observed during the pilot study which could lead to smoke borne illnesses. Through the use of the ICS, it gradually reduces the local people's exposure to high PM2.5 and high CO due to higher efficiency of combustion within their household over the lifetime of the project activity. The monitored baseline emission level from the open fire traditional cooking is 943.3 mg/MJ while the emission levels from newly distributed ICS is 177 mg/MJ.¹⁴ Monitoring Period: During this monitoring period, a total of 19,206 ICS has been newly distributed It should be noted that for ICSs that has been distributed for more than a year, there is a decrease in thermal efficiency and hence, there would be an increase in emissions of PM2.5 being released. This decrease in thermal efficiency is derived from the manufacturer of the ICS. However, during this current monitoring period, no ICSs were more than a year old as all ICSs were all newly distributed. Overall, there was a total decrease of 244.60 tonnes of PM2.5 from the usage of the ICSs. Additionally, this is further supported by the Monitoring Survey which reported a 100% of households have observed a decrease in intensity of the smoke.
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Impact #3	Less time spent on unpaid domestic and care work
Type of Impact	Positive, predicted, direct
Affected Stakeholder Group(s)	ICS end-users

¹⁴ http://catalog.cleancookstoves.org/stove_details.html?stove_id=stove_AWOUJ

Resulting Change in Well-being	Describe characteristics and magnitude of impact on each stakeholder group <u>Baseline:</u> Due to higher thermal efficiency, less firewood is required during the cooking process, and time spent cooking is significantly reduced. This results in a direct reduction in time spent on firewood collection and cooking for people. In the baseline, it was found that end-users would spend an average of 6.25% of a day collecting firewood. Upon sampling during the monitoring period, this proportion should reduce as less firewood is required for cooking and heating throughout the week. <u>Monitoring Period:</u> During this monitoring period, monitoring of the frequency of collection of firewood indicated that participating households now only spend 1.78% of a day collecting firewood on average, which is lower than the baseline. This decrease from the baseline scenario results in a total of 1.07 hours saved per day per participants during the monitoring period.
Impact #4	Increase Income
Type of Impact	Positive, actual, direct
Affected Stakeholder Group(s)	Local residents, Employees hired for Project Activity

Resulting Change in Well-being	Describe characteristics and magnitude of impact on each stakeholder group Baseline: During the project crediting period, TEM PNG Cookstoves Project Pte Ltd will oversee maintenance of ICSs and the monitoring plan, which will require the need to hire local people. Local people will be hired through a fair and unbiased process, during which we estimate that 130 jobs (10 full-time and 120 part-time) will be created over the project lifespan. The Project targets hiring females as at least 50% of all employees. Employees will be compensated above the average income of 439 Kina /month¹⁵. Monitoring Period: During this monitoring period, 24 local people were hired with 2 being full-time and 22 being part-time. Out of these 24 staff, 9 were female and 15 were male, resulting in a total of 37.50% being females. Employees who were employed part-time were compensated with an average income of 1,343.33 Kina/month.
Impact #5	Trainings imparted on climate change, project implementation and monitoring procedures
Type of Impact	Positive, actual, direct
Affected Stakeholder Group(s)	Local community, Beneficiaries, TEM PNG Cookstoves Project Pte Ltd in-country team and Staff

¹⁵ Bonilla, F., Voorend, K., Anker, R., Anker, M., & Prates, I. (2020). Anker Living Income Reference Value: Rural Papua New Guinea (2020) (No.20-03-20). Universidad Privada Boliviana.

Resulting Change in Well-being	Describe characteristics and magnitude of impact on each stakeholder group Baseline: Prior to the project activity, local residents and end-users had little awareness of ICSs and climate change. With the inception of the project activity, training and skill development related to community engagement, survey conduction, technical trainings like conducting Water Boiling Tests (WBT) will be provided to many stakeholder groups which is envisaged to empower their lives by not only providing long-term employability but also through increased awareness levels regarding issues related to climate change, social equity. This will be done during the onboarding of new staff throughout the lifespan of the project activity. Monitoring Period: During this monitoring period, local people who were hired were put through training and skill development as mentioned above. Training attendances and materials are present which can be made available to the validator upon request. A total of 24 local staff (2 full-time positions and 22 part-time positions) were trained for their respective roles. All staff are trained annually to ensure that knowledge transferred to them is up to date and the delivery of the project remains on high standards.
Impact #6	Improved Economic Outcomes
Type of Impact	Positive, actual, direct
Affected Stakeholder Group(s)	ICS end-users, ICS end-users' Families

Resulting Change in Well-being	Describe characteristics and magnitude of impact on each stakeholder group Baseline: With the commencement of the project activity, the use of the ICS increases fuel efficiency, and an increase in monetary savings is expected. This change is the result of the reduction in the cost of firewood purchases over the lifespan of the project activity due to the increase in fuel efficiency with the ICSs. As per the project activity's Emission Reduction Calculations for project activities in the Highlands, the quantity of firewood calculated to be saved was 3.70 tonnes/year. Coupled with the Baseline Survey Report where it was found that end-users typically spent K540/year on firewood, the potential savings an ICS end-user could save is 2,116.8 Kina or USD560.84. Monitoring Period: During this monitoring period, the project has resulted in savings of 3.63 - 3.70 tonnes of non-renewable biomass per year per ICS as well as a reduction in trips taken per week to collect or buy firewood. Additionally, it was recorded in the monitoring survey that approximately 5% of the end-users buy firewood (9% in Batch 1 and 2% in batch 2). The project has resulted in a total cost saved of 1.02 Kina per household per day in Batch 1 and 1.04 Kina per household per day in Batch 2. By proportioning the number of households that purchase their firewood from the monitoring survey, 57,211 Kina is saved during this period. Overall, the project ultimately results in long-term household savings during this monitoring period.
Impact #7	Reduction in sales of Firewood for Firewood Vendors
Type of Impact	Negative, actual, direct
Affected Stakeholder Group(s)	Firewood Vendors

Resulting Change in Well-being	Describe characteristics and magnitude of impact on each stakeholder group Baseline: In the absence of the project activity, end-users may opt to procure firewood in order to meet the volumetric need for daily cooking and heating practices. With the commencement of the project activity, there is a possibility that there will be a reduction in demand of firewood associated with the ICSs that might affect the vendors selling fuel wood in the project area. However, considering the significant demand-supply gap on fuelwood for cooking needs, decrease in the fuel-wood demand in project location is not likely to have a significant negative impact on wood vendors. Monitoring Period: During this monitoring period, it was determined that there was a decrease in fuel wood consumption from the baseline (file titled: TEM 2025_Cookstoves Highlands region report_v3.pdf) as mentioned in the decrease in cost spent by ICS end users in impact #6. The three-month monitoring period since the project began is too brief to detect meaningful changes in overall economic behavior. The majority of the rural population still rely on wood fuel, creating a demand-supply gap in the country. Hence, any reduction for wood-fuel within the project boundaries will shift stakeholders to sell wood outside the project boundary, without any significant reduction in their business.
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3.2 Stakeholder Impact Monitoring

During this monitoring period, TEM PNG Cookstoves Project's *trained* in-country team conducted sampling to collect data through interviews, surveys, direct observations, and group discussions regarding stakeholders' health and employment records.

The sampling plan for the SD VISta project activity has been designed based on the requirements of the allied VCS project activity – "Improved Cookstove Grouped Project in Papua New Guinea".

Project proponent will gather the necessary information to identify households using its ICS during the project. To facilitate this process, each ICS will be assigned a Unique Serial Number (USN). This number will be recorded during the registration process together with the following information (as appropriate and as available):

- Name of ICS user or head of the household
- Address / GPS of ICS household
- Phone number of ICS user or household, where available.

- ICS model
- Date of distribution/installation
- ICS serial number
- Retailer/distributor information

Additionally, sampling would have to be carried out for the project activity. The project activity distributes ICSs to households, which will have a net positive impact on the overall well-being of the people in the project area. To account for these positive impacts, the sampling plan identifies the parameters and sampling methods required.

The parameters monitored for the relevant activities with benefit for People and Prosperity and the respective evidence is mentioned in *Table 8* below.

Table 8: Parameters monitored for activities related to Benefits for People and Prosperity

S/N	Parameter	Unit	Description	Evidence Mode
1	Number of ICSs distributed	Number	Total number of ICSs distributed contributes to a reduction of firewood used, which is usually purchased at high prices.	Distribution Ledger
2	Level of fine particulate matter (PM2.5) emissions.	mg/MJ	Reduction in the level of fine particulate matter (PM2.5) emissions within households	Clean Stove Catalogue
			Reduction in the amount of smoke produced	Ground Monitoring Survey
			Reduction in thermal efficiency in older cookstoves	BURN study / Water Boiling Test as per VM0050 v1.0
3	Number of participants during training sessions	Number	Number of youth and adults in formal and non-formal education and training in the previous 12 months, by sex	Signed participation/ attendance forms, onboarding documents, employment contracts.
4			Number of youth and adults with technical and vocational skills, by type of skill within the project area.	

5	Employment contracts	Number	Number of people employed, by sector and sex, under the project activity	Project Employee Breakdown and Analysis Report
6	Employee Pay slips	Number	Average hourly earnings of employees, by sex, age, occupation, and persons with disabilities	Project Employee Breakdown and Analysis Report
7	Number of trips made to collect/buy firewood per week	Number	Proportion of time spent on unpaid domestic and care work, by sex, age, and location.	Ground Monitoring Survey
8	Number of women hired for the project activity	Number	Number of women in the project including managerial roles.	Female Employment contracts

3.3 Net Positive Stakeholder Well-being Impacts

The inception and distribution of ICSs undertaken by the project activity will provide a basic service necessary to lead to a healthy and productive life, including saving time and money for wood fuel at the household level. The project will reduce the drudgery undertaken by them especially for women and children through time savings in cutting, collecting, and carrying firewood from trees.

Through the development of the project activity, local stakeholders will yield the following benefits:

1. **19,206** households were provided with access to the fuel-efficient ICS for meeting their daily cooking needs.
2. Emissions of PM2.5 is reduced to be below the baseline emission level by **81.24%** to 177 mg/MJ in project households for ICSs. Older cookstoves provide a lower reduction. Overall, this results in a decrease in **244.60** tonnes of PM2.5 during this monitoring period. Additionally, this is further supported by Ground Monitoring Survey which reported a **100%** of households have observed a decrease in intensity of the smoke.
3. The proportion of time taken to collect firewood decreased from **6.25%** of a day in the baseline scenario to **1.78%** of a day during this monitoring period. This decrease resulted in a total time of **1.07 hours** saved per day per household.
4. An increase in the economic outlook and an improvement in the technical and vocational skills of local people as **24** local people were hired to be part of the project team. Out of

the people hired, 22 were hired on a part-time basis with an average income of **1,343.33** Kina per month.

5. There would be an increase of local people who are trained on climate change, project implementation and monitoring procedures. During this monitoring period, a total of **24** individuals were trained with **9** being female and **15** being male.
6. Families will have a greater capacity to save money used for buying wood fuel due to the reduced demand of fuel wood for cooking purposes. For end-users who buy firewood, this has resulted in total cost savings of 1.03 – 1.05 Kina per household per day. By proportioning the number of households that purchase their firewood from the monitoring survey, **57,211 Kina** is saved during this period in households that buy firewood.

4 BENEFITS FOR THE PLANET

4.1 Impacts on Natural Capital and Ecosystem Services

Impact #1	Reduced demand for large diameter non-renewable firewood from live trees due to only needing finger-sized woody biomass, i.e., twigs and crop residues
Type of Impact	Positive, actual, direct
Affected Natural Capital and/or Ecosystem Service(s)	Live plant biomass, Biodiversity and species Richness, Soil and Water Conservation
Resulting Change in Condition	Describe characteristics and magnitude of impact on each affected natural capital and ecosystem service Baseline: By slowing the consumption of woody biomass for cooking purposes and by eliminating the need for large diameter woody biomass due to the technology in the ICSs, the resulting drop in demand slows deforestation, thus increasing the above-ground biomass in the adjoining forests near the project area. Reduction in unsustainable harvesting of plant biomass would in turn reduce soil erosion and result in improvement in biodiversity and species richness of the implementation area. Monitoring Period: During this monitoring period, both the frequency of firewood collection and the amount spent on firewood has decreased. This drop in demand would slow down deforestation and thus increasing the above-ground biomass in the adjoining forests near the project area The project has resulted in savings of tonnes of non-renewable biomass per year per ICS. Over this monitoring period, a total savings of 13,446 tonnes of biomass from the 19,206 ICS distribution.
Impact #2	Avoided emission of GHGs made possible by using ICSs
Type of Impact	Positive, predicted, indirect

Affected Natural Capital and/or Ecosystem Service(s)	Air quality in the households as well as surroundings
Resulting Change in Condition	Describe characteristics and magnitude of impact on each affected natural capital and ecosystem service <u>Baseline:</u> The ICSs require less fuelwood for producing the same amount of thermal energy, therefore, resulting in less carbon dioxide emission in the atmosphere. As carbon dioxide is a greenhouse gas, its reduced emission translates to a positive impact on climate change. An annual average of 320,099 tCO₂ is estimated to be avoided due to the inception of the project activity. <u>Monitoring Period:</u> During this monitoring period, a total of 13,443 tCO₂ was estimated to be reduced due to the inception of the project activity.

4.2 Natural Capital and Ecosystem Services Impact Monitoring

Similarly to section 3.2, the impact on natural capital and ecosystem services were monitored by TEM PNG Cookstoves Project in-country team. The data was collected through surveys and kitchen performance tests (KPTs). The data from tests and surveys were recorded, analysed and reported by TEM PNG Cookstoves Project. The parameters monitored for the impacts relevant to benefits for the planet with its respective evidence during this monitoring period are summarised in *Table 9* below.

Table 9: Monitoring plan for activities with benefit for the Planet

S/N	Parameter	Unit	Description	Evidence Mode
1	Number of ICSs distributed during the monitoring period	Number	Total number of ICS distributed contributes to a reduction of firewood used, which is usually purchased at high prices.	Distribution Ledger
2	Tonnes of CO2 removed during the monitoring period	TCO2	Tonnes of greenhouse gas emissions avoided or removed	Kitchen Performance Tests (KPTs) & Baseline Surveys & Emission Reduction Calculation Sheet
3	Tonnes of biomass saved during the monitoring period	Tonnes	Progress towards sustainable forest management through the increase of above-ground biomass in forests	

4.3 Net Positive Natural Capital and Ecosystem Services Impacts

By replacing traditional low efficient traditional open fire cooking with high efficiency ICSs in households, the project will increase energy efficiency resulting in less firewood combustion, thus generate net GHG reductions.

The average annual GHG emissions reduction from the project is expected to be **320,099 tCO₂e**. The crediting period is expected to be 10 years. Hence the total GHG emissions reduction is **3,200,995 tCO₂e**. The project will save 3.63 - **3.70 tonnes** of non-renewable biomass every year and it will result in a significant reduction of deforestation.

For this monitoring period, the GHG emissions reduced were **13,443 tCO₂e** and a total of **13,446 tonnes** of non-renewable biomass were saved.

APPENDIX I: STAKEHOLDER CONSULTATIONS

During the current monitoring period, the following stakeholder interactions at the SHP and WHP were conducted:

Ongoing consultation	Non-formal education and training was provided to local people to be converted to local staff
Date(s) of stakeholder consultation	08 July 2024 – 11 July 2024
Communication of monitored results	Training was provided by TEM staff in the local language. Training modules included: 1) Introduction to ICSs Community Engagement & Communication 2) ICSs Health & Safety Field Team Roles 3) Data Collection, tools, and Hands on Practice 4) Health, Safety and Security for the Project An informal quiz is conducted at the end of the training course to test the local staff's knowledge. The passing rate of the quiz is at least 70%, and passing is compulsory.
Consultation records	Training records of all training courses are kept, and certificates are presented to staff who have completed their training.
Stakeholder input	All staff trained are made known that they can always seek clarification from the TEM PNG office in the event that they have any questions on training materials or company policies. Any input on training will be received and the training process will be re-evaluated if needed.
Ongoing consultation	Awareness and Household Registration of houses to take part in the ICSs program.
Date(s) of stakeholder consultation	19 August 2024 – 31 October 2024

Communication of monitored results	Prior to the distribution of ICSs, awareness will be conducted on the benefits of the ICS in every ward. Willing households taking part in the project would be required to register with their particulars and household information being recorded. During this, households would be registered and given a token to exchange for an ICS.
Consultation records	Information was recorded in documents by software to keep track of the ownership of the ICS. A total of 19,206 household registrations were done.
Stakeholder input	Stakeholders gave information such as household's geographical location, contact details of the head of the family or end-users and consent forms to be filled by interested end-users.

Ongoing consultation	Collection of ICS by end-users who have undergone Household Registration previously
Date(s) of stakeholder consultation	23 September 2024 – 31 October 2024
Communication of monitored results	ICS end-users would bring tokens to the distribution sites where they would be collected by the distribution team. Collected tokens would be exchanged with an ICS.
Consultation records	A data entry officer would be present to manage data collected by the registration team. A total of 19,206 tokens were collected and exchanged with an ICS.
Stakeholder input	All information provided was collected and managed by the data entry officer and the warehouse and logistic officer who manages the data collated from the households. Information of end-users would be updated once they have received the ICS. Information collected will include: • Name of ICS user,

	• Address/GPS of ICS household • Phone number of ICS user • ICS model • Date of distribution and installation • Name of staff that delivered and installed the ICS • Photographic Evidence
Ongoing consultation	A day of training was conducted for TEM's in-country field assistants before conducting the survey. This is followed by two days of ground monitoring surveys to be conducted for this monitoring period for users of the ICS as per the sampling plan.
Date(s) of stakeholder consultation	15-Jan-2025 to 17-Jan-2025
Communication of monitored results	Communication would be conducted in their native language by TEM's in-country team. End-users would be made clear on the reason for this survey.
Consultation records	Results from the ground monitoring survey would be collected and collated to be used in the ER calculation. A total of 120 households were selected, and 96 Sampling Surveys were successfully completed.
Stakeholder input	All inputs which are required by the sampling plan will be collected from end-users through measurements, observations and a list of questions.